

The empirical recurrence for OEIS A230907, recovered from its tail

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Abstract

OEIS A230907 records “Empirical recurrence of order 89” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 359$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 95$. Its last coefficient is $c_{89} = 128 \neq 0$, so no further tail satisfies anything shorter; any order-89 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A230907 is “Number of $n \times 5$ 0..3 arrays $x(i,j)$ with each element horizontally, vertically or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 4$, no adjacent elements equal, and upper left element zero.”. It has offset 1 and begins

0, 84, 634, 4308, 30304, 219696, 1603356, 11727974, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:05:15, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 359$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 359$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 89: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 89.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 718$ exact terms from $n = 6$ on, it returns order 89 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	40	c_{24}	−611136019	c_{47}	−15810757316	c_{70}	−28502643
c_2	−748	c_{25}	732767505	c_{48}	5840603633	c_{71}	−9280734
c_3	8735	c_{26}	−636479410	c_{49}	16514589224	c_{72}	15741335
c_4	−71587	c_{27}	1039895833	c_{50}	2567165436	c_{73}	377816
c_5	437527	c_{28}	−1861924996	c_{51}	−12002446647	c_{74}	−9998975
c_6	−2063406	c_{29}	1842534616	c_{52}	−2670015323	c_{75}	14260227
c_7	7642985	c_{30}	−392321179	c_{53}	618820221	c_{76}	−5387501
c_8	−22338059	c_{31}	−2457150377	c_{54}	3771827388	c_{77}	−1184826
c_9	51010036	c_{32}	6233993273	c_{55}	1295146669	c_{78}	847000
c_{10}	−88022065	c_{33}	−6896877642	c_{56}	−2125988841	c_{79}	−545384
c_{11}	104559556	c_{34}	855229615	c_{57}	2660077040	c_{80}	535862
c_{12}	−56852247	c_{35}	4604960472	c_{58}	−1112884568	c_{81}	−230322
c_{13}	−64306121	c_{36}	−2685849750	c_{59}	−2723036110	c_{82}	169162
c_{14}	184875169	c_{37}	4329495575	c_{60}	724850435	c_{83}	−135000
c_{15}	−203793002	c_{38}	−11010079113	c_{61}	696414956	c_{84}	37520
c_{16}	150634291	c_{39}	257251987	c_{62}	1195623344	c_{85}	3656
c_{17}	−201002790	c_{40}	18223250731	c_{63}	−104134003	c_{86}	−4248
c_{18}	391443653	c_{41}	−6936259947	c_{64}	−1216253823	c_{87}	1952
c_{19}	−437814657	c_{42}	−11524580065	c_{65}	235229235	c_{88}	−768
c_{20}	134225808	c_{43}	−3920629111	c_{66}	149267081	c_{89}	128
c_{21}	221898830	c_{44}	11800112438	c_{67}	−13580851		
c_{22}	−296815278	c_{45}	16275318017	c_{68}	161463430		
c_{23}	341259047	c_{46}	−16030008395	c_{69}	−80983625		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 95$, and it is the recurrence OEIS A230907 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{89} - c_1x^{89-1} - \dots - c_{89}$. Its constant term is $-c_{89} = -128$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 89 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 89, so $\deg q = 89$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 89, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 89, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 128.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-89 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A230907.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.